Name: Parth Pandurang Kale
SRN: PES1UG24AM184
Section: D

Problem Statement #26 — Peer-to-Peer Carpooling & Expense Splitter

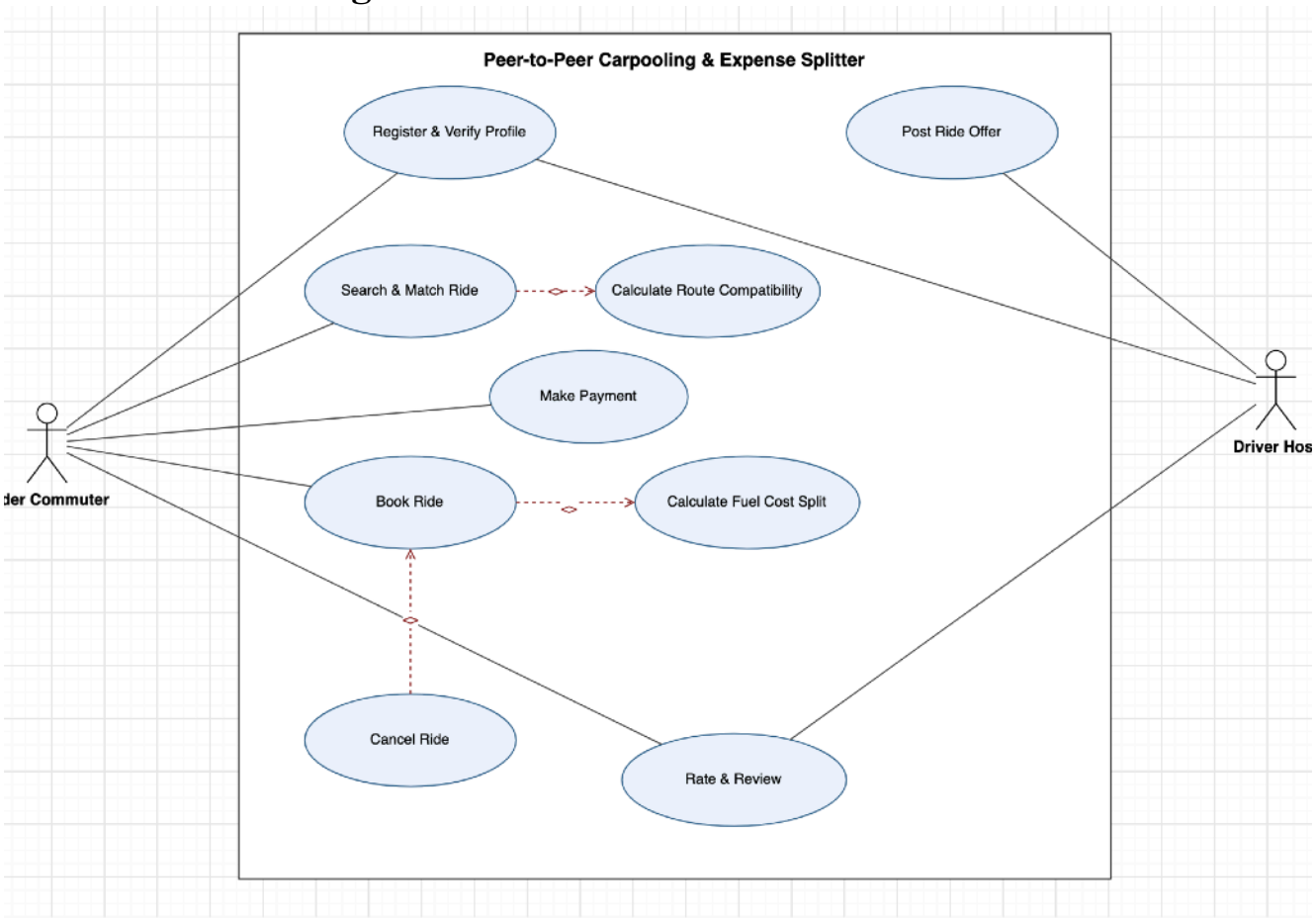
Functional Requirements:

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall calculate route compatibility between a driver's and rider's commute paths and permit a match only when the resulting detour is within a 1.5 km tolerance threshold.	High	Pass: Route match is confirmed and displayed only when detour ≤ 1.5 km. Fail: A match is confirmed when detour exceeds 1.5 km.	Keeps matches practical for drivers and prevents excessive detours that would discourage participation.
FR-002	The system shall automatically calculate each participant's fuel cost share for a matched ride, splitting the estimated fuel cost proportionally to the distance travelled by each rider.	High	Pass: Cost split is generated and displayed to all participants before ride confirmation and sums to 100% of trip fuel cost. Fail: Split is missing, inaccurate, or does not sum correctly.	Automated, transparent cost-sharing is the core value proposition of the expense-splitter feature.
FR-003	The system shall require Driver Host users to upload a valid driving licence and vehicle registration document, and shall block ride-posting until these are verified.	High	Pass: 'Post Ride Offer' is disabled until documents are marked verified. Fail: A driver can post a ride offer with unverified/missing documents.	Verifying drivers protects rider safety and is a baseline trust requirement for a carpooling marketplace.
FR-004	The system shall allow a Rider Commuter to book a matched ride and shall notify the Driver Host, who can accept or reject the booking request.	High	Pass: Booking request reaches the driver and ride status updates to 'Confirmed' or 'Rejected' based on driver response. Fail: Booking is auto-confirmed without driver action, or status does not update.	A two-sided confirmation step ensures both parties consent to sharing the trip.
FR-005	The system shall allow either party to cancel a confirmed ride up to a configurable cutoff time before departure and shall notify the other party immediately.	Medium	Pass: Cancellation before cutoff updates ride status to 'Cancelled' and sends a notification within 1 minute. Fail: Cancellation is silently accepted with no notification, or is allowed after the cutoff without flagging a late cancellation.	Reliable cancellation handling prevents no-shows and preserves trust between commuters.

Non-Functional Requirements:

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Security & Privacy	User ratings, driving-licence verification documents, and travel histories must be stored securely, encrypted at rest and in transit, with encrypted backups.	High	Pass: Security audit / penetration test confirms encryption (AES-256 at rest, TLS 1.2+ in transit) and no plaintext PII in logs or backups. Fail: Any sensitive field is stored or transmitted unencrypted.	Licence documents and travel history are sensitive personal data; a breach would cause real-world safety and legal exposure.
NFR-002	Performance & Scalability	The route-matching and cost-splitting engine shall return results within 2 seconds under a simulated peak load of 10,000 concurrent match requests during morning/evening commute windows.	High	Pass: Load-test report shows 95th-percentile response time ≤ 2s at 10,000 concurrent requests with 0% error rate. Fail: Response time exceeds 2s or error rate is non-zero under the same load.	Carpooling demand is concentrated in short commute windows, so the system must stay responsive exactly when usage peaks.

UML Use-Case Diagram



Notes

- Actors: Rider Commuter and Driver Host, both associated with the system boundary 'Peer-to-Peer Carpooling & Expense Splitter'.
- «include» relationship: 'Search & Match Ride' always includes 'Calculate Route Compatibility'; 'Book Ride' always includes 'Calculate Fuel Cost Split' — both are mandatory sub-behaviours factored out of the base use case.
- «extend» relationship: 'Cancel Ride' extends 'Book Ride' — it is an optional behaviour that only occurs under the condition that a confirmed ride is being cancelled.
- 'Rate & Review' and 'Register & Verify Profile' are shared use cases invoked by both actors after a completed trip / during onboarding respectively.

Use-Case Flow Specification

Use Case Name	Book Ride
Use Case ID	UC-05
Primary Actor	Rider Commuter
Secondary Actor	Driver Host
Related Includes	Calculate Route Compatibility, Calculate Fuel Cost Split
Related Extends	Cancel Ride (extends this use case)
Preconditions	• Rider Commuter is registered, logged in, and has a verified profile. • At least one Driver Host has an active, verified ride offer posted for an overlapping route/time window.
Postconditions	Success: Ride status is 'Confirmed'; a seat is reserved; both parties hold matching trip and fuel-cost-split records. Failure: Ride status remains 'Pending' or reverts to 'Available'; no seat is reserved; no cost-split record is created.
Main Success Scenario	1. Rider Commuter searches for a ride by entering origin, destination, and desired travel time. 2. System executes the <<include>> use case 'Calculate Route Compatibility' and returns Driver Host offers whose detour is within the 1.5 km tolerance. 3. Rider Commuter selects a matching ride offer and requests to book it. 4. System executes the <<include>> use case 'Calculate Fuel Cost Split' and displays the rider's estimated cost share. 5. Rider Commuter confirms the booking request. 6. System notifies the Driver Host of the pending booking request. 7. Driver Host reviews and accepts the request. 8. System updates ride status to 'Confirmed', reserves the seat, and sends confirmation notifications with trip and cost-split details to both parties.
Alternate Flow — Driver Rejects Booking	a. At step 7, Driver Host rejects the booking request. b. System updates ride status to 'Rejected' and notifies the Rider Commuter that the seat is unavailable. c. System returns the Rider Commuter to the list of matching ride offers from step 2 so they may select an alternative. d. Use case ends without a confirmed booking.